

Introduction

Shohei Ohtani is an MLB player for the Los Angeles Dodgers and formerly for the Los Angeles Angels. He is unique in that he both bats and pitches, something rare in professional baseball. Even more rare, he is elite at both. In 2022, he finished 4th in Cy Young voting (the best pitcher award) and 2nd in MVP voting (the best player award). He is a 4x MVP winner, having won in 2021 and 2023-2025. He has won the Silver Slugger Award (best hitter at his position) 4x. He also won rookie of the year. He has revolutionized baseball by being such a so-called unicorn; he is truly a once-in-a-generation type of player.

The question I am attempting to answer is whether Shohei hits differently when he's pitching than when he's only batting. I chose this topic because I'm a sports and baseball fan. I'm interested in what makes a player great, and what factors can affect a great player, which naturally led me to this question. Going into this project, I think that there will not be a difference because the effects of pitching fatigue will balance out with being more focused and engaged with the game, something that MLB players say has a huge impact on their performance (see Rafael Dever's breakup with the Red Sox over not being able to play in the field, an effect I'd imagine would only be increased by pitching due to the even more greater focus needed).

To analyze Shohei's performance on a game-by-game basis, I'll use a stat called wOBA (weighted on-base average). wOBA is defined by the equation

$$\text{wOBA} = \frac{0.69 \cdot \text{NIBB} + 0.722 \cdot \text{HBP} + 0.881 \cdot 1\text{B} + 1.257 \cdot 2\text{B} + 1.593 \cdot 3\text{B} + 2.058 \cdot \text{HR}}{\text{AB} + \text{BB} - \text{IBB} + \text{SF} + \text{HBP}}.$$

If you're not familiar with wOBA, it's not overly important. You can study the chart below, which shows a good rule of thumb for assessing wOBA.

wOBA Rules of Thumb	
Rating	wOBA
Excellent	.400
Great	.370
Above Average	.340
Average	.320
Below Average	.310
Poor	.300
Awful	.290

The data I am using is every regular-season game Shohei played from 2021 to 2023 in which he had at least 3 plate appearances. I chose these years because it was the start of his prime and before he injured his arm, which prevented him from pitching in the entire 2024 season and much of the 2025 season. I treat each game as an observation from a bigger,

theoretical population of possible games that describe his true offensive ability during this period. Essentially, the population is the outcomes that would occur had Shohei played an unlimited number of games in this time period. Thus, the games from 2021 to 2023 are a sample from this population, meaning we can infer his true offensive ability with a large sample size.

To analyze the data, I'll generate the wOBA on a per-game basis, then define $\bar{x}_p$ as the mean wOBA of games in which Shohei pitched, and $\bar{x}_{np}$ as the mean wOBA of games in which Shohei did not pitch. Then, I'll perform a 2-sample t-test and a 95% t-interval for the difference of population means to infer underlying ability. I am using a t-test and t-interval because I am working with means and the population standard deviation is unknown. My parameters of interest are

$$\begin{aligned}\mu_p &= \text{true mean game wOBA when pitching} \\ \mu_{np} &= \text{true mean game wOBA when not pitching.}\end{aligned}$$

My hypotheses are

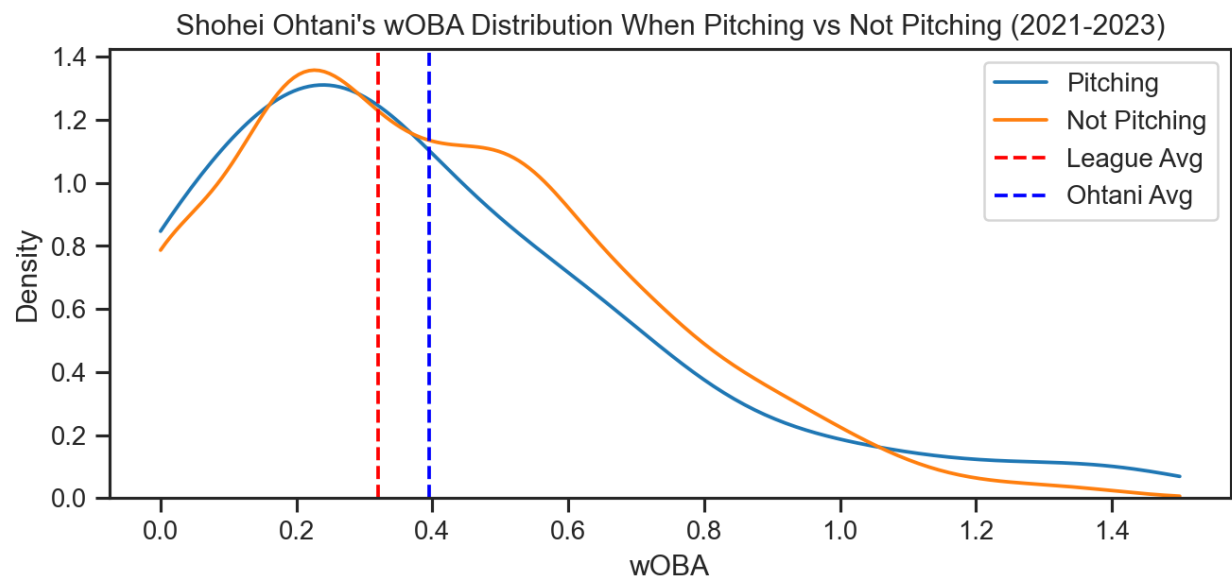
$$\begin{aligned}H_0 &: \mu_p = \mu_{np} \\ H_a &: \mu_p \neq \mu_{np}.\end{aligned}$$

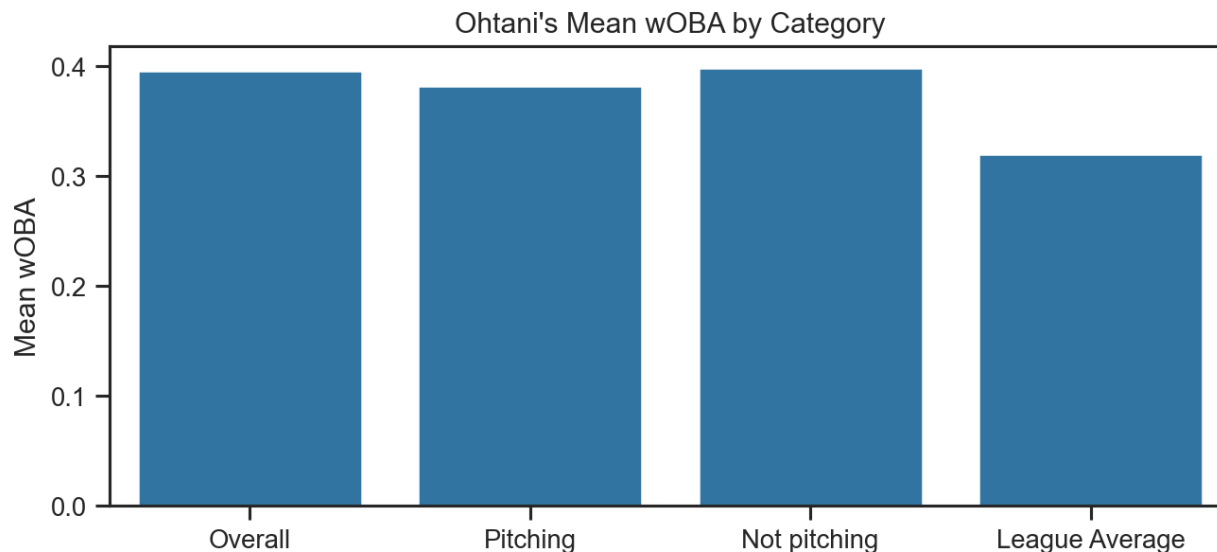
Plan

I obtained my data from [Baseball Reference](#). I went to Shohei’s game logs and copied the batting game logs for each year from 2021 to 2023. I created the unintentional walks column by subtracting intentional walks from intentional walks, and the singles column with $1B = H - (2B + 3B + HR)$. I used the formula to calculate wOBA from this. These game logs contain a position column, which I used to distinguish between games he pitched in and didn’t. Since multiple positions can be played in a game (for example, in a start, he’d exit the game after a certain number of innings, and would play DH for the rest of the games), I split the position on each space and found which lists contained “P”. I then split the dataset into two datasets, one in which Shohei pitched and one in which he didn’t.

Statistic	Pitching	Not Pitching	Overall
Count	68	363	431
Mean	0.381	0.398	0.395
Std. Dev.	0.335	0.290	0.297
Min	0.000	0.000	0.000
25%	0.175	0.174	0.174
Median	0.294	0.352	0.345
75%	0.565	0.568	0.566
Max	1.422	1.372	1.422

wOBA summary statistics





The mean wOBA is 0.398 when not pitching compared to 0.381 when pitching, and the median is 0.352 versus 0.294, showing slightly higher values when not pitching. His mean wOBA regardless of position is 0.395, which is 0.075 higher than the league average wOBA (a substantial increase, as the standard deviation is roughly .040). This indicates that Shohei does in fact perform better when not pitching. It also shows that both distributions are right skewed, as the mean is greater than the median for each, which supports the density plot. The standard deviation for pitching is higher than when not pitching ($s_p = 0.335 > s_{np} = 0.280$), as is the range ($1.422 > 1.372$), indicating higher variability. This can likely be explained by the smaller sample size ($n_p = 68 < n_{np} = 363$). **So, it appears that while Shohei performed slightly better when pitching compared to when not pitching, it is likely not statistically significant**, due to the smaller sample size when pitching and the fact that the standard deviations for each is much bigger than the difference in means. H_0 is unlikely to be rejected.

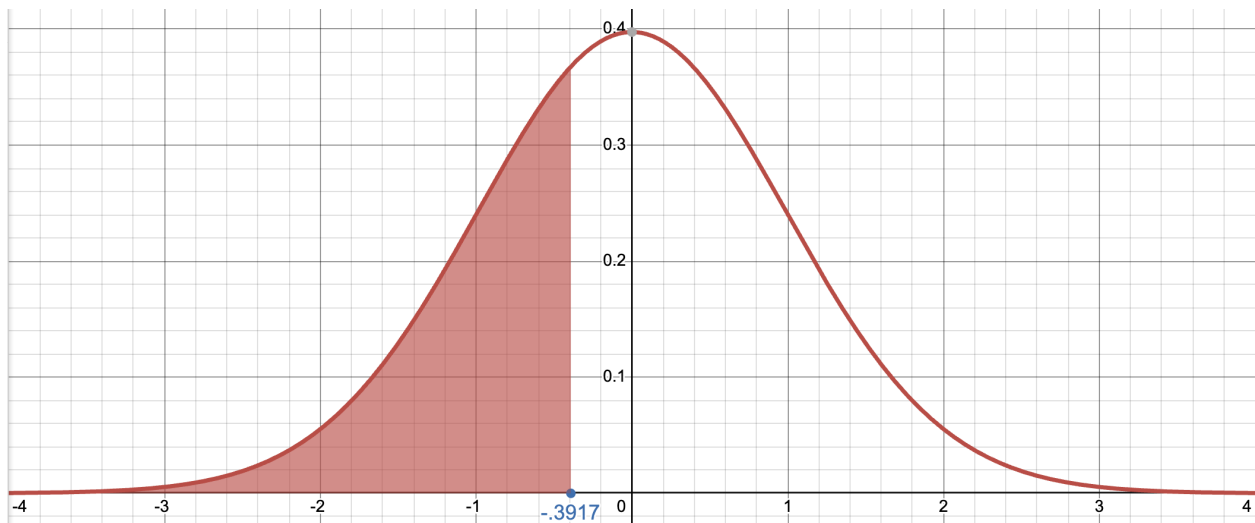
We'll test this, but first, the conditions for inference must be met. First, the pitching data is a random sample from the previously discussed larger, theoretical population of all games Shohei could have pitched in; the not pitching data is also a random sample from the population of all games that Shohei could have batted but not pitched in. Second, since this population is theoretically unlimited, the 10% conditions are met for both ($\frac{1}{10}\infty = \infty > n_p = 68$ and $n_{np} = 363$). Last, the CLT applies to both, as $n_p = 68 \geq 30$ and $n_{np} = 363 \geq 30$, meaning both sampling distributions are approximately normal. Therefore, the conditions for inference are met.

Analysis

Now, we'll perform the two-sample t-test for $\mu_p - \mu_{np}$.

$$\begin{aligned} t &= \frac{\bar{x}_p - \bar{x}_{np}}{\sqrt{\frac{s_p^2}{n_p} + \frac{s_{np}^2}{n_{np}}}} \\ &= \frac{0.381 - 0.398}{\sqrt{\frac{(0.335)^2}{68} + \frac{(0.290)^2}{361}}} \\ &= \frac{-0.017}{0.043397} \\ &= -0.3917 \end{aligned}$$

$$\begin{aligned} p &= 2 \cdot \text{tcdf}\left(\overset{\text{lower}}{-1\text{E}99}, \overset{\text{upper}}{-0.3917}, \overset{\text{df}}{67}\right) \\ &= 2 \cdot 0.3483 \\ &= \boxed{0.6966} \end{aligned}$$



Assuming H_0 is true (Shohei's true mean wOBA is the same when pitching vs not pitching), there is a 0.6966 probability of getting a sample with a mean difference as large or larger than the one in this sample (.017).

Since $p = 0.6966 > \alpha = 0.05$, we fail to reject H_0 . There is not convincing evidence that Shohei Ohtani's true mean wOBA is different when pitching vs not pitching. This conclusion would hold for any common value of α (0.01, 0.05, 0.10). This supports my prediction that Shohei would bat similarly regardless of position.

There is a possible type II error, because we failed to reject H_0 . This would mean that Shohei actually does bat differently when pitching compared to not pitching. Although a type I error is not possible, as we did not reject H_0 , it would mean that Shohei does not bat differently, but we said that he did.

Next, we'll perform a two-sample t-interval for $\mu_p - \mu_{np}$, at confidence level 95%.

$$\begin{aligned}
 \text{CI} &= (\bar{x}_p - \bar{x}_{np}) \pm t^* \sqrt{\frac{s_p^2}{n_p} + \frac{s_{np}^2}{n_{np}}} \\
 &= (\bar{x}_p - \bar{x}_{np}) \pm \text{invT}\left(1 - \alpha/2, \text{df}\right) \sqrt{\frac{s_p^2}{n_p} + \frac{s_{np}^2}{n_{np}}} \\
 &= (0.381 - 0.398) \pm \text{invT}\left(0.975, 67\right) \sqrt{\frac{(0.335)^2}{68} + \frac{(0.290)^2}{361}} \\
 &= -0.017 \pm 1.996 \cdot 0.043397 \\
 &= -0.017 \pm 0.08662 \\
 &= \boxed{(-0.1036, 0.0696)}
 \end{aligned}$$

We are 95% confident that $(-0.1036, 0.0696)$ contains the true mean difference in wOBA when Shohei is pitching vs not pitching.

Since 0 is on the interval, $H_0 : \mu_p = \mu_{np}$ is plausible. This supports the t-test, in which we failed to reject H_0 . The interval gives us a range of plausible values for $\mu_p - \mu_{np}$, showing there is likely a small to zero difference when pitching vs not pitching.

Conclusion

Both the t-test and t-interval imply that there is no real difference in Shohei Ohtani's batting ability when pitching vs not pitching. Thus, Ohtani should pitch whenever he can. There is likely no reason to prevent him from pitching in order to improve his batting performance or vice versa. However, an experiment would be needed to be run to confirm this result, as cause and effect cannot be inferred, because this is an observational study rather than an experiment.

This is one limitation of my conclusions. Another is that the sample size for pitching was much smaller than when not pitching, which increases the variability and the chance of a type II error. I did not control for weather, opponent, rest days, injuries, etc., which could effect the conclusion. Game-to-game wOBA is noisy, and I did not control for the number of plate appearances, besides only using games with at least 3. A more accurate way to run the tests would be to weight the wOBA by plate appearances, although the math would get much more complicated and would likely produce similar results.

The study could be improved by increasing the sample size, meaning it would have to be repeated in the future after Ohtani had played more games. It could also be improved by accounting for variables like weather and rest, but a more advanced statistical method, such as regression, would need to be used, rather than a t-test.

Overall, the results suggest that Shohei's batting is consistent regardless of whether or not he is pitching. While there is no statistically significant difference, the data and methods I used have limitations, meaning a more complete and extensive study may be able to identify a difference. Nevertheless, the findings highlight how special of a player Shohei is, as he is able to hit at an elite level, even while simultaneously performing of the most physically demanding roles in all of sports.